

Ing. Šimon Hudínek

Process Control Engineer  portfolio-3d8.pages.dev

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Process control engineer covering the whole cycle: data acquisition, modeling, control design, and actuation. My thesis runs a reaction-kinetics reactor model inside a nonlinear MPC. At Škoda Auto I instrumented a dilution tunnel and built the tool that automatically picks the right dilution ratio (1st place, SVK 2025).

Experience

R&D Intern, Simulation & Modeling, Škoda Auto *Emission Laboratory (EPS/3) · 03–08/2025*

- Modelled humidity transport through the dilution tunnel of an emission test bench in Python, from mass and energy balances, predicting the conditions under which condensation invalidates a test run.
- Instrumented the tunnel with humidity and temperature sensors along its length and validated the model against that campaign.
- Delivered an application that evaluates every available dilution ratio for a given vehicle, from its mass, road-load coefficients and driving cycle, and selects the lowest ratio that still holds a margin against condensation.

Researcher & Data Analyst, Monolith Research Group, UCT Prague *2022–2025*

- Extended the group's 1D Diesel Particulate Filter model to resolve the soot deposition profile along the filter wall, which the previous model did not represent, introducing a flow-heterogeneity parameter (χ) calibrated against 3D CFD so the overall filtration efficiency is preserved. Bachelor's thesis.
- Restored the group's laboratory test rig after a measurement-card failure: remapped the wiring to the replacement hardware and rebuilt the LabVIEW control logic.

Projects

Predictive AdBlue Dosing in Diesel Aftertreatment (Master's Thesis) *2025–2026*

- Nonlinear MPC for AdBlue injection in diesel-generator aftertreatment: too little dosing lets NO_x escape, too much causes ammonia slip. The controller model runs on the reaction kinetics themselves, so it tracks the ammonia stored on the catalyst washcoat, a state no sensor can measure.
- Turned XMR, an in-house Fortran reactor simulator, into a controller component: compiled it into a callable library so the controller can run several warm-started instances per step. Average solve time 5.8 s against a 30 s control step, on the development machine.
- Built the cost function on top, drove it with IPOPT, and scripted the weight tuning. The NMPC previews load steps 8 minutes ahead and holds NH₃ slip below 1 ppm, enough to remove the downstream ammonia-slip catalyst, at the cost of some NO_x conversion at partial load; urea consumption is 5.7 % lower than the PID baseline.

Multi-Platform Water Control System (team semester project) *09–12/2025*

- Led the engineering side of an 8-person team (programmers, testers, frontend) delivering a control system for a physical multi-tank water plant on schedule.
- Designed the control system myself: first-principles model linearized around the operating point, linear MPC (do-mpc) with EKF state estimation, NI-DAQmx acquisition, and a PySide6 operator app, achieving stable closed-loop control on the real rig.
- Rebuilt the same plant as a distributed architecture in software: three OPC UA server nodes in place of field controllers, supervisory MPC above them, a web HMI, and a data historian.

Embedded Systems (personal projects)

- ESP32 circadian lighting cabinet run by a local server on the home network: firmware, electronics, and enclosure, in daily use. ESP32 smart bottle: capacitive water-level sensing, Firebase logging, offline buffering and power management.

Education

MSc, Sensorics and Cybernetics in Chemistry *UCT Prague, 2024–2026*

BSc, Nano and Microtechnology in Chemical Engineering *UCT Prague, 2021–2024*

- BSc thesis: Efficient 1D Models of Filtration in Catalytic Filters for Air Protection. Coursework in process control, system identification, and numerical simulation. Grade average 1.16 in the MSc and 1.61 in the BSc.

Awards

SVK student research conference: **1st place 2025** (Škoda CVS humidity tool), **2nd place 2023** (DPF filtration modeling) Škoda Auto AI Hackathon 2025: **4th place**

Technical Skills

Modeling & Control: first-principles modeling (thermodynamics, fluid dynamics, reaction kinetics), system identification, MPC (linear and nonlinear), nonlinear optimization (IPOPT), state estimation (EKF)

Software: Python (NumPy, SciPy, PySide6, Tkinter), MATLAB/Simulink, Fortran, C/C++, Git, Linux

Field & Lab: sensor instrumentation, NI-DAQmx, LabVIEW, electronics, OPC UA, test-rig troubleshooting

Languages: Czech (native), English (C1), German (A2)

Additional: EU citizen, available for relocation and work-related travel, driving licence B